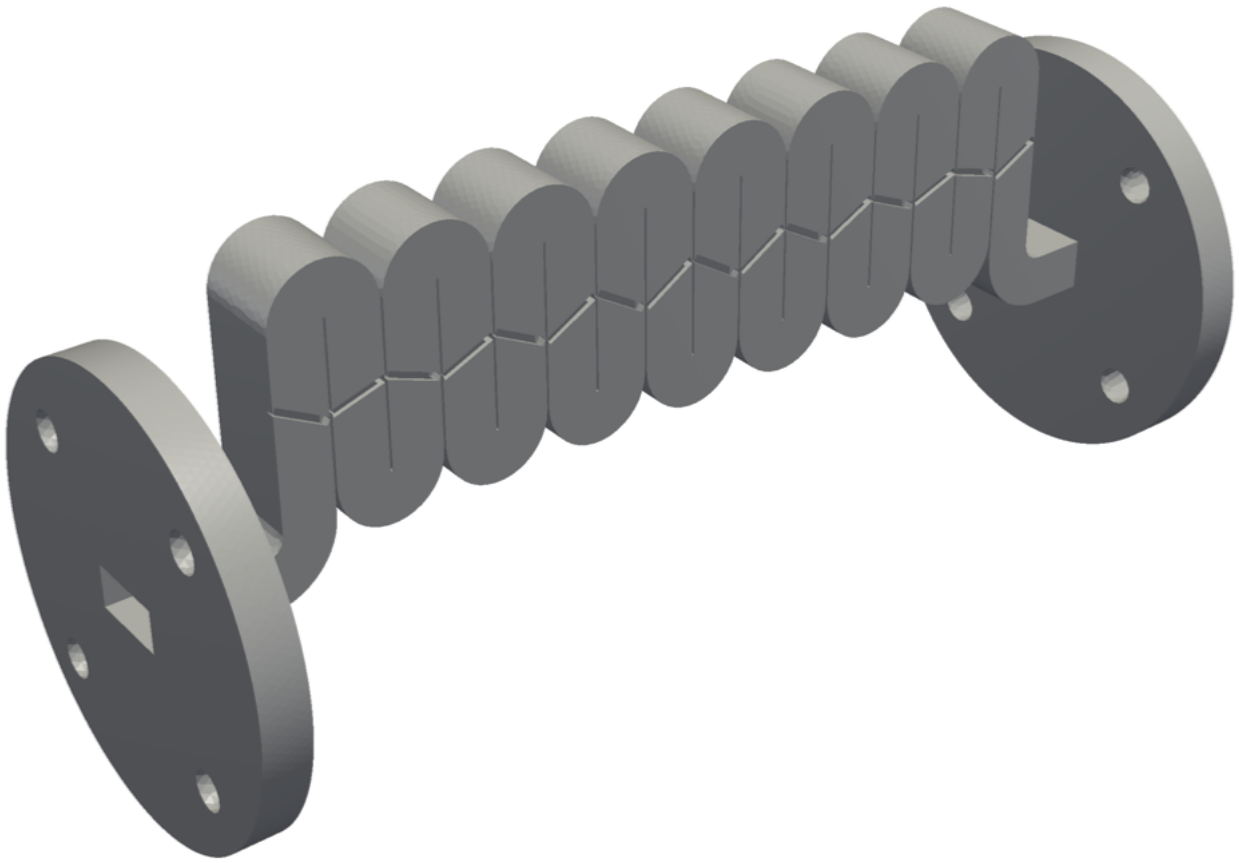
60-GHz Serpentine Leaky-Wave Antenna

A brief from basics to advanced, and a replica you can print



The replica built in this repository from the paper's equations and Table II (STEP / STL).

Paper	Result
J. Gabsteiger, C. Robisch, A. Koelpin, F. Lurz, "Advancing 60-GHz SISO Radar: Additively Manufactured Leaky Wave Antenna With Enhanced Beam Steering", IEEE Trans. Antennas Propag., vol. 74, no. 7, pp. 6130–6145, July 2026, DOI 10.1109/TAP.2026.3681348 (CC BY 4.0)	**76°** beam steering over **58–63.5 GHz** (5.5 GHz) with two ports · 16 slots · **13–15 dBi** · 3D-printed resin, silver-coated · ≈ 1° angle accuracy in a SISO radar

How to read this: every section is tagged **BASIC**, **INTERMEDIATE** or **ADVANCED**. Read only the green parts for the idea, add the orange parts for how it works, and the red parts to design or build one.

BASIC

1. The big idea in one page

A radar finds **how far** a target is easily, but finding **in which direction** it is usually needs many receive antennas and channels (MIMO) or a motor that turns the antenna. This paper does it with **one transmit and one receive channel**: the antenna itself points its beam in a different direction for every frequency. When the radar sweeps its frequency (a chirp), the beam sweeps across the room like a searchlight, and the moment a target answers tells the angle.

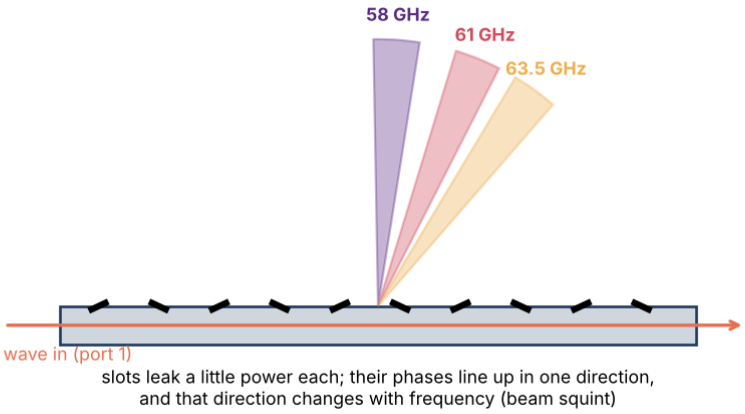


Figure 1: a leaky-wave antenna. A wave runs inside a metal tube (waveguide); small slots let a bit of it leak out. The direction where all leaks add up depends on frequency.

Everyday picture	In the antenna
A garden hose with small holes: water leaks out of every hole	a waveguide with 16 slots: microwaves leak out of every slot
Waves from all holes add up in one direction, like a stadium 'wave'	the main beam, about 8° wide
Change the water pressure and the spray angle changes	change the frequency and the beam angle changes (≈ 6° per GHz)
Fold the hose into a zig-zag to make the effect stronger	the serpentine: a 4.6× longer path between slots, so 4× more steering

Why 60 GHz? The 57–64 GHz band is licence-free and has several GHz of bandwidth. Ready-made radar chips exist (Infineon BGT60TR13, used here), and the antennas are only a few cm long. More bandwidth means both finer range and more beam steering.

BASIC

2. How the radar measures range and angle

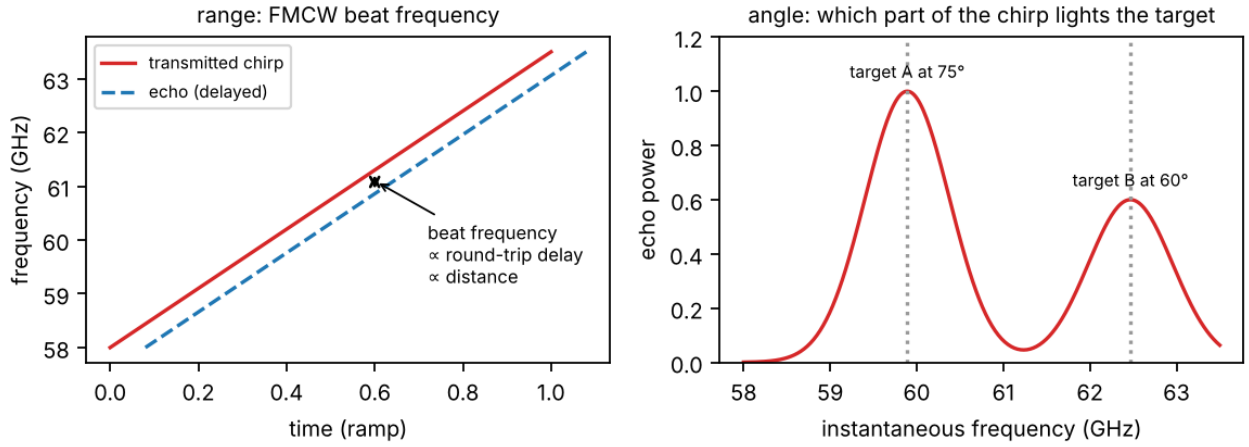


Figure 2: left, FMCW range. The echo is a delayed copy of the chirp, and their difference (the beat frequency) grows with distance. Right, the angle. A target only answers while the beam points at it, i.e. during a certain part of the chirp.

INTERMEDIATE

The formulas

$$\Delta R = c / (2B) \text{ (eq. 1)} \quad R = c \cdot f_{beat} \cdot t_{mod} / (2B) \text{ (eq. 3)}$$

With B = 5.5 GHz, ΔR = 2.7 cm. But a squinting beam only illuminates a target for part of the chirp, so only a slice B' of the bandwidth carries that target's range information:

$$\Delta R = c / (2B') = (c / 2B) \cdot \theta_s / \theta_{3dB} \text{ (eq. 2)} \quad \Delta R \cdot \theta_{3dB} = c \cdot \theta_s / (2B) \text{ (eq. 4)}$$

Equation 4 is an **uncertainty relation**: a narrower beam (better angle) means fewer GHz per target (worse range). For our design $\theta_s \approx 32°$ per port and $\theta_{3dB} \approx 7.7°$, so each beam position gets B' ≈ 1310 MHz, and ΔR ≈ 11 cm. The paper quotes 30 cm for a 500 MHz slice.

Signal processing

The received beat signal is cut into overlapping windows (a short-time Fourier transform, STFT). Each window belongs to a known frequency slice, hence a known beam direction, and its FFT gives the range of the targets in that direction. This is the same maths as FFT beamforming, only the 'spatial' axis is replaced by the frequency–angle law of the antenna.

> **Dual port + switch:** feeding the antenna from the other end mirrors the beam about broadside. A waveguide switch alternates the two ports, so 5.5 GHz of physical bandwidth gives an 11 GHz 'virtual' sweep and 76° of coverage.

INTERMEDIATE

3. Waveguides and slots

A rectangular waveguide is a hollow metal pipe. Above its cutoff frequency it carries the TE10 mode, whose electric field spans the short side b and varies along the long side a. For 60 GHz the standard size is **WR15: a = 3.76 mm, b = 1.88 mm**, cutoff $f_c = c/2a = 39.9$ GHz. The wave inside travels with phase constant

$$\beta = 2\pi / \lambda_g = \sqrt{((2\pi f)^2 \mu_0 \varepsilon_0 - (\pi/a)^2)} \quad \text{(eq. 6)}$$

β is always smaller than the free-space $k = 2\pi/\lambda_0$, and it grows with frequency faster than k does. That difference is what makes a leaky-wave antenna steer.

Why tilted slots in the narrow wall?

Currents on the narrow wall flow across it, around the guide. A slot cut straight across the narrow wall runs parallel to those currents and does not interrupt them, so it does not radiate. Tilting it by θ_n cuts the currents, and the radiated power grows roughly with $\sin^2\theta_n$. The slot is longer than b, so it wraps around the corners into both broad walls to a depth d_n; together, tilt and depth set the slot's **conductance** g_n, the fraction of power it takes out. Neighbouring slots are tilted in opposite directions, which adds 180° of phase between them.

Standing-wave vs travelling-wave arrays

	Standing-wave ($d = \lambda_g/2$)	Travelling-wave ($d \neq \lambda_g/2$): this paper
phase between slots	exactly 180° (+180° from the tilt) → in phase	frequency-dependent
beam	fixed at broadside, narrow band	scans with frequency (leaky-wave)
end of guide	short circuit	matched / second port

The beam direction (Floquet theory)

A periodic array radiates where the phases of all elements line up (eq. 7), $k\,d\cos\varphi - \beta\,d = 2\pi\,m$. Solving for the angle gives eq. 8. Only the $m = -1$ space harmonic radiates (eq. 9), and the alternating slot tilt doubles the period to 2d:

$$\varphi = \arccos(\beta/k - \lambda / (2d)) \quad \text{(eq. 10)}$$

Straight slotted guide with d = 2.38 mm: 111.1° at 58 GHz and 102.3° at 63.5 GHz, **only 8.7° of steering**. Not enough for a radar that needs at least 40°.

INTERMEDIATE

4. The serpentine trick

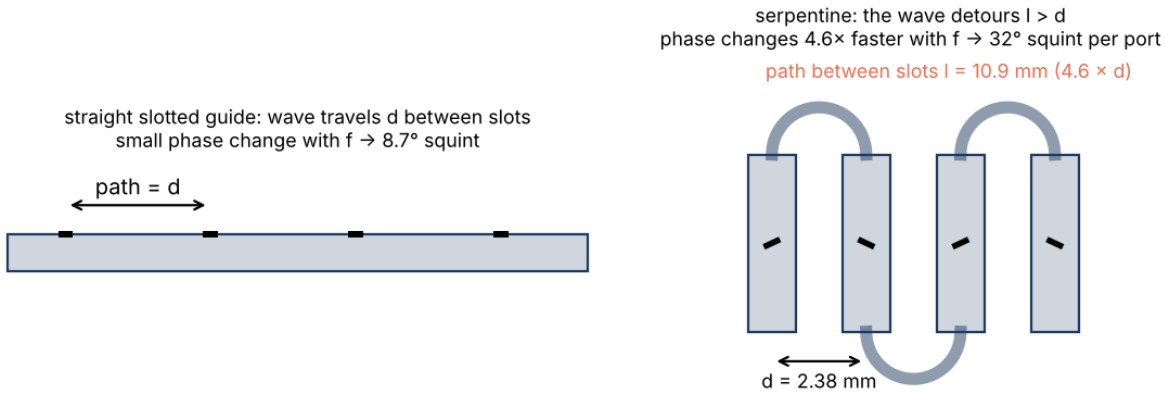


Figure 3: the slots stay 2.38 mm apart, but the wave detours through a U-bend between them.

If the wave travels a path **l** between slots that are **d** apart, the phase between slots is βl (not βd). A small change of β with frequency then turns into a large change of phase, so the beam moves much further:

$$\theta_0 = \arccos\left((\beta\, l - 2\pi(m + \tfrac{1}{2})) / (k\, d) \right) \quad \text{(eq. 11)}$$

Design steps from the paper, reproduced exactly by `02_python/lwa_design.py`:

Step	Rule	Paper	This replica
slot spacing d	$d < \lambda_0/2$ at 63.5 GHz (no grating lobe); d = b + 0.5 mm	2.38 mm (limit 2.36)	2.38 mm (limit 2.36)
path length l	θ_0(58 GHz) = 86°, m = 1	10.9 mm	10.903 mm
beam at 58 GHz	eq. 11	86°	86.05°
beam at 63.5 GHz	eq. 11	53.8°	53.89°
scan per port		32.2° (sim.), 34° (meas.)	32.2°
with port 2	mirror about 90°	76° total	72° between beam peaks (+ beamwidth ≈ 76°)

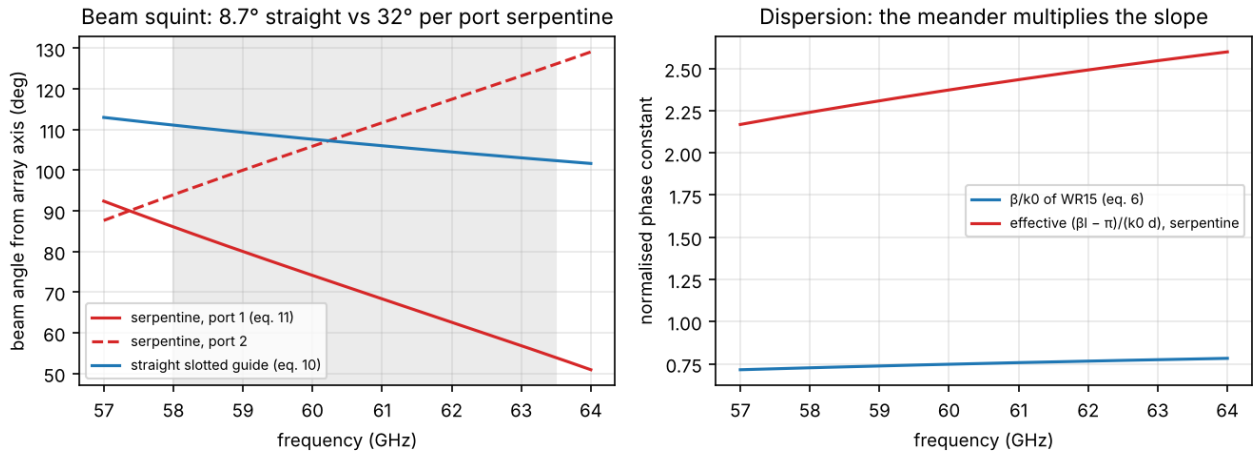


Figure 4: beam angle and effective dispersion, computed with this repository's design script.

ADVANCED

5. Designing the 16 slots

Each slot takes a share of the power that is still travelling in the guide. To get low sidelobes, the radiated amplitudes follow a **Dolph–Chebyshev** taper (N = 16, 20 dB sidelobes). For a given sidelobe level it gives the narrowest possible main beam:

$$(AF)_{2M} = \Sigma\, a_n \cos[(2n-1)\,\psi/2],\ \psi = k\,d\cos\varphi - \beta\,l \quad \text{(eq. 15)} \quad g_n = a_n^2 / \Sigma\, a_i^2 \quad \text{(eq. 16)}$$

The paper turns each g_n into a tilt and a depth with polynomial fits of CST unit-cell runs (eq. 17, 18; the logarithm is the natural log, checked against Table I). The first model radiated too little (S21 ≈ −3 dB), so all tilts were scaled by 1.5, depths recomputed with eq. 20, and the slot widened from 0.261 mm (eq. 19, w = a/(16·0.9)) to 0.34 mm for printing. That gave S21 ≈ −6.5 dB and S11 < −15 dB.

n	a_n ours / paper	g_n ours / paper	tilt ours / paper (°)	$tilt_{adj}$ paper (°)	$depth_{adj}$ paper (mm)
1	0.867 / 0.866	0.0696 / 0.069	18.77 / 18.731	28.097	0.195
2	0.504 / 0.504	0.0235 / 0.023	11.45 / 11.332	16.998	0.308
3	0.622 / 0.621	0.0358 / 0.035	13.73 / 13.656	20.484	0.281
4	0.733 / 0.733	0.0498 / 0.049	16.00 / 15.947	23.921	0.248
5	0.833 / 0.832	0.0642 / 0.064	18.05 / 18.017	27.026	0.21
6	0.914 / 0.913	0.0773 / 0.077	19.74 / 19.71	29.566	0.173
7	0.971 / 0.97	0.0872 / 0.087	20.93 / 20.907	31.361	0.145
8	1.0 / 1.0	0.0926 / 0.092	21.55 / 21.526	32.289	0.13

Slots 9–16 mirror slots 8–1. The tilt formula eq. 17 is reproduced to within 0.15°. The printed depth fit (eq. 18) is rounded in the paper and gives depths up to 0.09 mm off, so the CAD uses the paper's own Table II values.

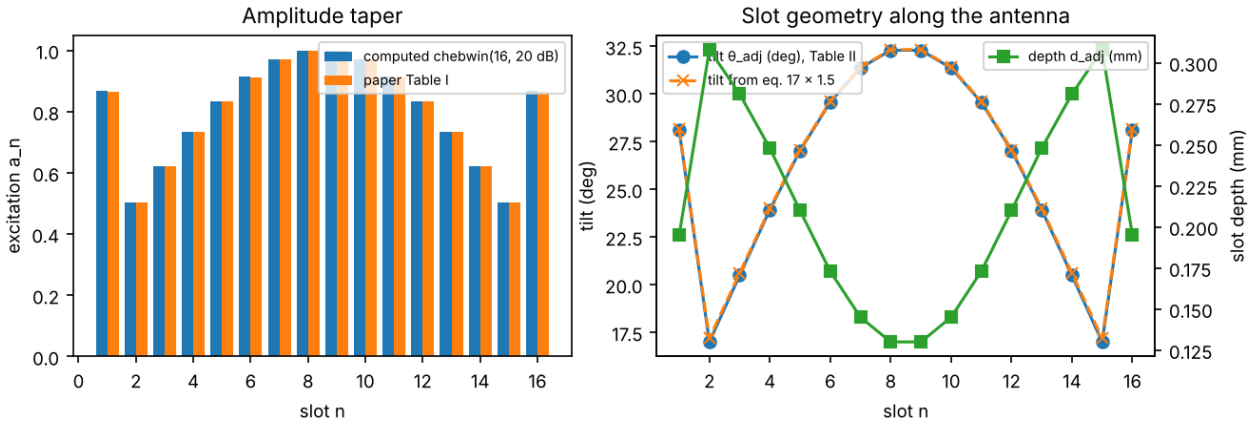


Figure 5: taper and slot geometry along the antenna.

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6. Radiation patterns, beam map and the open stopband

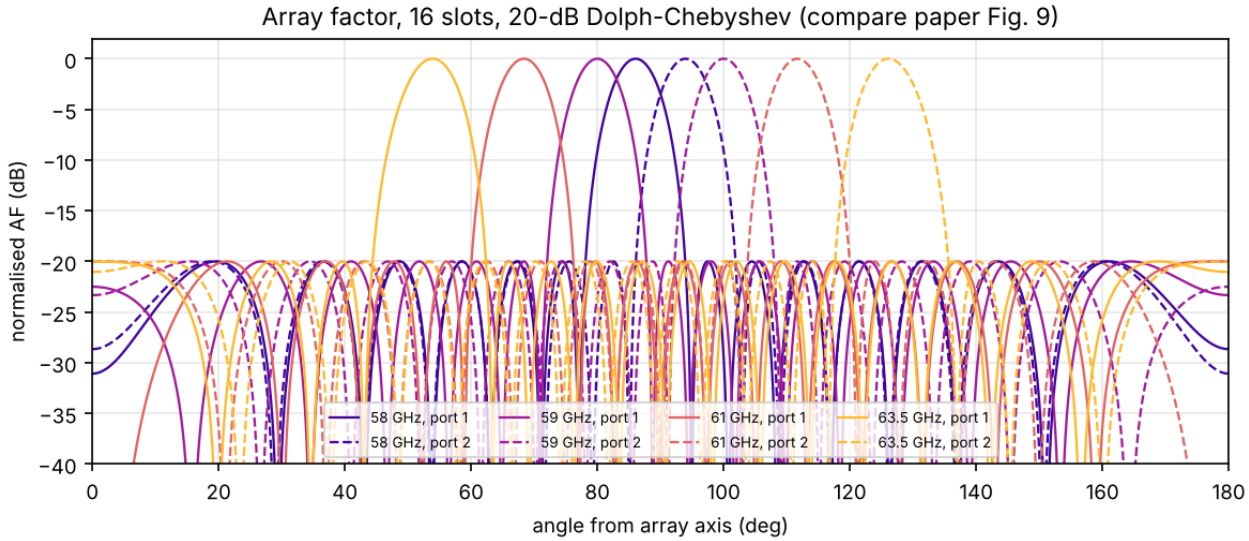
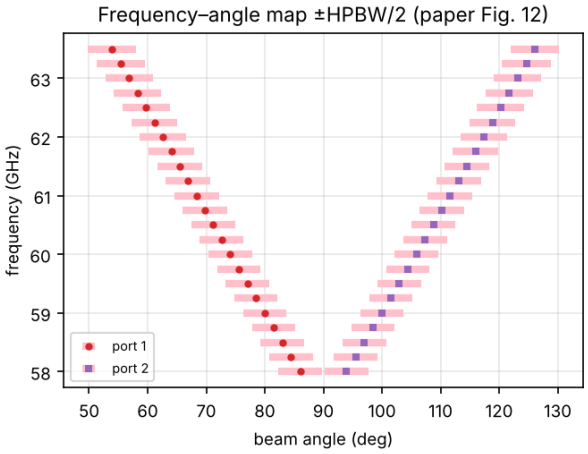


Figure 6: array factor of the 16 tapered slots for both ports (compare with paper Fig. 9). The real antenna adds the slot element pattern and the gain dip near broadside.



Frequency–angle map (paper Figs. 12/26): each port covers one side of broadside, almost linearly, at about 5.8° per GHz in this model (7°/GHz measured). The bars show the half-power beamwidth.

Open stopband (OSB). Near broadside ($\beta_{eff} \to 0$, around 57.5–59 GHz) the forward and backward space harmonics couple, all reflections add in phase, and S11 rises to ≈ −2 dB while the gain dips. It is usually removed by breaking the symmetry of the unit cell. The paper tried this (Fig. 10/11) but kept the symmetric design because both ports must behave the same; the horn receiver partly hides the dip.

Polarisation: 12.5 dBi co-pol and 2.5 dBi cross-pol, so XPD ≈ 10 dB (eq. 21), which is enough for radar.

ADVANCED

7. How it is made: SLA printing and silver plating

Step	Paper's process	Why it matters
Printer	Formlabs Form 3+, clear resin, 25 µm layers	smooth walls; a 0.34 mm slot needs fine resolution
Orientation	tilted 5–12° to the build plate (Fig. 17)	flat causes suction-cup peeling and micro-cracks; steep (> 20°) causes staircase edges
Cleaning	Form Wash, then isopropanol + compressed air through the guide	resin stays trapped in the narrow slots and bends
Cure	UV in Form Cure	dimensional stability
Activation	$SnCl_2$ + HCl + water dip	creates seeds so silver can grow on plastic
Silvering (Tollens')	1.5 g $AgNO_3$ in 75 mL water, 0.3 g NaOH, NH_4OH until clear, then 1.5 g dextrose in 30 mL water; 1–5 min, 3 cycles	a conformal metal layer with no electrical contact needed (fume hood, PPE)
Pump-assisted	reagent circulated through the guide by a peristaltic pump	removes deposits in the bends that shortened the path l
Result	Ag 14–22 µm thick; the skin depth at 60 GHz is ≈ 0.25 µm	more than 50 skin depths, so the loss is not limited by plating thickness
Shrink fix	printed opening was 3.38 × 1.72 mm: widen by 100 µm, lengthen each half loop by 200 µm	the beam angle depends on l and a, so 0.2 mm errors shift it by degrees

Sensitivity in numbers: the first prototype, without the pump, steered only 47° and was shifted by about 16°. The corrected, pump-plated one reached 76° and matched the simulation within 1–2° (rms gain error ≈ 1 dB).

8. Measured performance and comparison

Work	Squint	Rel. bandwidth	Band
[18] (sim. only)	48°	34.3 %	28–40 GHz
[19]	7°	7.4 %	91–98 GHz
[20]	17°	18.8 %	24–29 GHz
[21] metal FDM print	19°	24.4 %	9–11.5 GHz
[22] serpentine	30.6°	37.5 %	12–18 GHz
[23]	39°	6 %	9.7–10.3 GHz
[10] same group, parallel-plate	42°	16.6 %	55–65 GHz
this paper	**76°**	**9.1 %**	**58–63.5 GHz**

Measured: gain 13–15 dBi, 76° with both ports, angle test 9° measured for a 10° target, i.e. 1° error.

ADVANCED

9. The replica in this repository

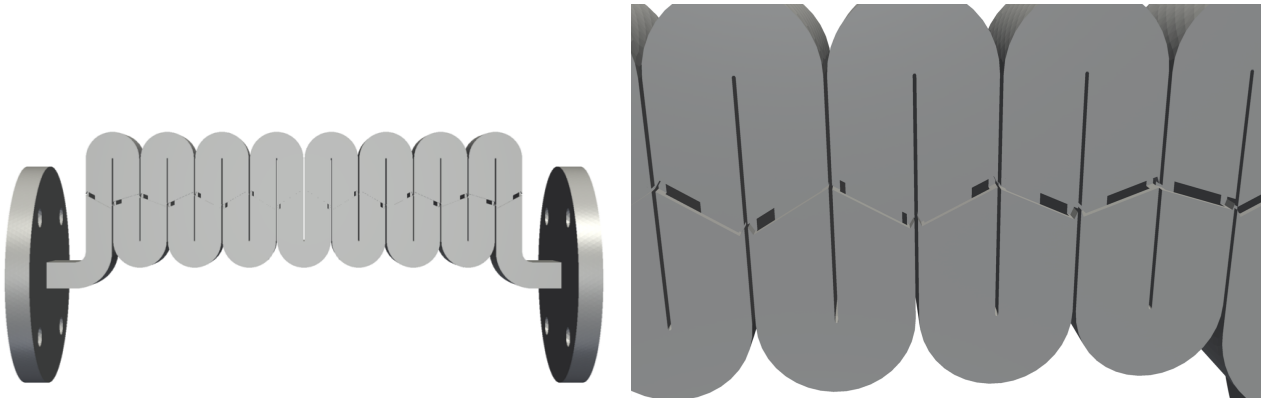


Figure 7: replica CAD. Left: the radiating face with the serpentine legs and two flanges. Right: close-up of the alternately tilted slots, each cut through the narrow wall and into the broad walls.

Item	Value	Source
waveguide a × b	3.76 × 1.88 mm (WR15)	paper
slot pitch d	2.38 mm	paper eq. 13
wall t / air gap	0.2 / 0.1 mm	assumed (t + t + gap = 0.5 mm)
path between slots l	10.9 mm; straight leg 7.162 mm + semicircular bend Rc = 1.19 mm	paper l; bend shape assumed
slots	16, width 0.34 mm, tilt/depth from Table II, alternating ±tilt	paper
feeds	both at the bottom of legs 1 and 16, 90° E-plane elbow (R = 2 mm), horizontal WR15	assumed
flanges	Ø19.05 mm, 4 × Ø1.7 mm holes on Ø14.29 mm, 44.8 mm apart	span from paper; flange pattern simplified, check against your UG-385/U
files	01_cad/lwa_shell.step, lwa_shell.stl (print), lwa_air.step (FEM air volume)	

To build one

1) Print `lwa_shell.stl` on an SLA printer (25 µm), tilted 5–12°, with supports away from the slot face. 2) Measure the printed opening and adjust `P['a'], P['b'], P['l']` in `build_lwa_cad.py` to compensate shrinkage (the paper needed +0.1 mm and +0.2 mm). 3) Clean, cure, activate and silver-plate with the pump (section 7). 4) Measure S11/S21 on a VNA with WR15 transitions and the pattern on a turntable. **The 0.1 mm air gap between legs is at the limit of desktop SLA**; if it fuses, the slot ends see metal instead of air and the conductances change, so re-tune the depths.

ADVANCED

10. Full-wave check of one slot (Hertzon / EMerge)

`03_sim/slot_unit_cell.py` rebuilds the paper's unit cell in the open-source FEM simulator Hertzon (EMerge solver). It is a straight WR15 with one inclined slot (w = 0.261 mm) and two wave ports, radiating into an air box with an absorbing boundary. The radiated fraction $P_{rad} = 1 - |S11|^2 - |S21|^2$ gives the slot conductance through $P_{rad} = 4g / (2 + g)^2$.

tilt (°)	depth (mm)	f (GHz)	S21 (dB)	S11 (dB)	P_{rad}	g sim	g paper Fig. 4	tets
10.0	0.38	58.0	-0.04726	-45.22	0.01079	0.01091		33146
10.0	0.38	60.75	-0.02906	-49.33	0.00666	0.0067	0.018	33146
10.0	0.38	63.5	-0.01871	-53.25	0.00429	0.00431		33146
20.0	0.29	58.0	-0.2543	-30.47	0.05598	0.05935		41145
20.0	0.29	60.75	-0.1532	-33.76	0.03424	0.03547	0.075	41145
20.0	0.29	63.5	-0.09301	-37.01	0.02099	0.02144		41145
30.0	0.2	58.0	-0.6293	-22.64	0.12945	0.14953		55441
30.0	0.2	60.75	-0.3575	-25.93	0.07647	0.08294	0.16	55441
30.0	0.2	63.5	-0.2056	-29.17	0.04503	0.04718		55441

Result: conductance rises steeply with tilt (≈ $\sin^2\theta$) and S11 stays small, as the paper's method assumes. At 60.75 GHz the values are about 0.4–0.5× the paper's Fig. 4; at 58 GHz they come within 20–35 % for the 20° and 30° slots. So this cell's slot resonance sits lower than the paper's. Likely reasons: the exact depth definition, the open air beside the broad walls instead of the 0.1 mm gap and neighbouring leg, and the coupling to neighbouring slots in the paper's CST cell. The mesh is converged (10° slot: 5.5k → 75k tets changes P_{rad} from 0.0062 to 0.0067). For a build, re-tune depths with this script until each slot reaches its g_n at the centre frequency.

What a full model would need

The complete antenna is about 45 × 12 × 4 mm with 0.34 mm slots and thin walls. A converged FEM run is roughly 1–3 million unknowns per frequency. That is feasible on a 32–64 GB workstation with Hertzon, CST or HFSS using `01_cad/lwa_air.step`, but not on the 4-core / 15 GB machine used here.

BASIC

11. Glossary

Term	Meaning
LWA	leaky-wave antenna: a guided wave that radiates continuously along its length
beam squint	the beam direction changes with frequency
SISO / MIMO	single / multiple input, multiple output: number of radar channels
FMCW	frequency-modulated continuous wave radar: a chirp; range comes from the beat frequency
STFT	short-time Fourier transform: FFTs of short overlapping windows
WR15	standard waveguide for 50–75 GHz, 3.76 × 1.88 mm
TE10, cutoff	the basic waveguide mode; below cutoff (39.9 GHz for WR15) nothing propagates
β, λg	phase constant and wavelength of the wave inside the guide
Floquet harmonic	the space harmonics of a periodic structure; m = −1 is the one that radiates
grating lobe	an unwanted second main beam when elements are spaced more than ≈ λ/2 apart
Dolph–Chebyshev	amplitude taper with equal sidelobes and the narrowest beam for a given sidelobe level
conductance g	normalised power a slot takes out of the guide
HPBW	half-power (−3 dB) beamwidth
OSB	open stopband: mismatch and gain dip near broadside in periodic LWAs
XPD	cross-polarisation discrimination, co-pol / cross-pol power
SLA	stereolithography 3D printing: a laser cures liquid resin layer by layer
Tollens' reaction	chemical silver deposition from a silver-ammonia complex and a sugar
skin depth	depth to which RF current flows in a metal; 0.25 µm for silver at 60 GHz

Files: 02_python/lwa_design.py (all design equations, CSV and figures) · 01_cad/build_lwa_cad.py (STEP / STL) · 03_sim/slot_unit_cell.py (FEM check) · 04_docs/make_brief.py (this PDF).